

DAVID ARTHURIAN

Tujunga, CA 91042 | (818) 642-0100 | davidwarthurian@ieee.org | [LinkedIn](#)

EXPERIENCE

Northrop Grumman

May 2025 - Present

Electrical Engineer

- Serve as Cognizant Engineer for the Electronic Power Subsystem (EPS) on NASA's HALO spacecraft, leading design, integration, and verification of power distribution hardware supporting critical spaceflight operations.
- Develop, analyze, and validate power electronics and PCB-level hardware, ensuring high-reliability operation in radiation and thermal environments.
- Collaborate with cross-functional teams (systems, mechanical, software, and suppliers) to drive requirements, root-cause hardware issues, and deliver mission-critical hardware under aggressive schedules.
- Conduct circuit-level simulations (LTSpice, OrCAD, Simulink) and hardware bring-up/debug using oscilloscopes, power analyzers, and automated test scripts (Python/Matlab).
- Lead failure analysis and corrective actions for hardware anomalies, improving reliability and reducing integration risk.
- Author and review technical documentation, schematics, and test procedures to meet NASA and industry standards.

Sempra Energy

June 2023 - May 2025

Systems Engineer

- Collaborated with an engineering team to design and optimize Southern California CNG/LNG distribution systems for a large-scale critical infrastructure network serving over 500,000 residential, commercial, and industrial customers.
- Developed real-time monitoring and analytics tools using SCADA, GIS, SAP, and SQL to enhance system performance.
- Created emergency response dashboards for system engineers, ensuring rapid diagnostics and failure mitigation.
- Led instrumentation and control system projects, integrating hardware and software for high-reliability gas distribution.
- Conducted SQL-based data modeling to identify system improvements that increased distribution efficiency by 50%.
- Compiled data, created dashboards for emergency scenarios, and provided on-call support for region engineers.
- Designed and executed six-figure capital projects, achieving 25% cost savings and delivery ahead of schedule.

PROJECTS

Embedded Motion-Control System | Arduino, IMU Sensors, Python, Unity

Jan. 2024 - May 2024

- Designed and built an interactive, sensor-based multiplayer video game with IoT communication.
- Created hardware-based motion controllers using Arduinos, IMU sensors, and LEDs for gesture-based input.
- Embedded real-time embedded control and wireless communication protocols for multiplayer synchronization.
- Engineered the game logic in Python and developed a Unity-based graphical interface for stable and fun gameplay.

Line Following Car Competition - 1st Place

April 2023 - May 2023

- Programmed and assembled an autonomous robotic vehicle for maze navigation using PID controllers and IR sensors.
- Integrated real-time embedded control algorithms in C++, optimizing performance for speed and accuracy.
- Achieved 1st place out of 30 participants, completing the challenge with a record time of sub 10 seconds.

RF Signal Detection System | Python, RTL-SDR, FFT, DSP

July 2022

- Built a real-time RF sensing pipeline processing live IQ samples through FFT spectral analysis, dynamic noise-floor thresholding, and signal clustering to detect and characterize persistent RF activity. Designed as a modular software reference model for a future FPGA-accelerated DSP engine.

SKILLS

Python/MATLAB, Asyncio, C++, SQL, Verilog, VHDL, JavaScript, LTSpice, OrCAD, Altium, PCB Design, Simulink, Oscilloscopes, Power Analyzers, IBM DOORS, Cameo, Creo, Windchill, Git, VSCode, SCADA, SAP, GIS

EDUCATION

The University of California, Los Angeles

B.S. Electrical Engineering | GPA 3.75 | IEEE Member, HKN Society

EIT Certificate in Electrical Engineering

#182187